Introduction

As a cybersecurity enthusiast, I believe hands-on practice is crucial to mastering network security concepts. In this project, I set up a virtual lab using **Kali Linux** on VMware to simulate **ARP spoofing** and **DNS spoofing** attacks. This practical demonstration highlights how attackers manipulate network traffic and offers insights into how such attacks can be detected and mitigated.

This project is not only a learning experience but also a contribution to raising awareness about the vulnerabilities in network communication and the importance of securing DNS infrastructure.

What is DNS & ARP Spoofing?

DNS Spoofing is the attack where the victim receives an incorrect query to the requested DNS from the threat actor to steal information or implement a malware into the target device (Tunggal 2025).

ARP Spoofing is a Man In the Middle Attack where the threat actor interferes communication between the network devices (Imperva ca. 2025).

Objectives

The goal of this project was to:

1. Simulate a **man-in-the-middle (MITM)** attack by redirecting a target device's traffic using **ARP spoofing**.
2. Perform **DNS spoofing** to redirect requests for a legitimate domain (e.g., tiktok.com) to a spoofed IP address.
3. Use **Wireshark** to monitor and verify the attack in action.

Prerequisites:

Here's what you'll need to replicate this project:

- **Kali Linux** (configured in a virtual environment using VMware or VirtualBox)
- **Bettercap** (pre-installed on Kali Linux)
- **Wireshark** (to capture and analyze network traffic)
- A **target device** (e.g., a smartphone connected to the same network)
- A personal or isolated network environment (e.g., a mobile hotspot or home router)

Methodology

1.1 **Setting Up the Lab**

a) Install Kali Linux on VMware and configure it to use the bridged network mode to interact with devices on the same network.
b) Connect your phone or other target device to the same network (e.g., a personal hotspot).
c) Update Kali Linux and Install bettercap with the following commands
   - ***Sudo apt update***
   - ***Sudo apt install bettercap***

1.2 **Enabling IP Forwarding**

Open up the root terminal and type in the following commands to enable IP forwarding as shown in figure 1.

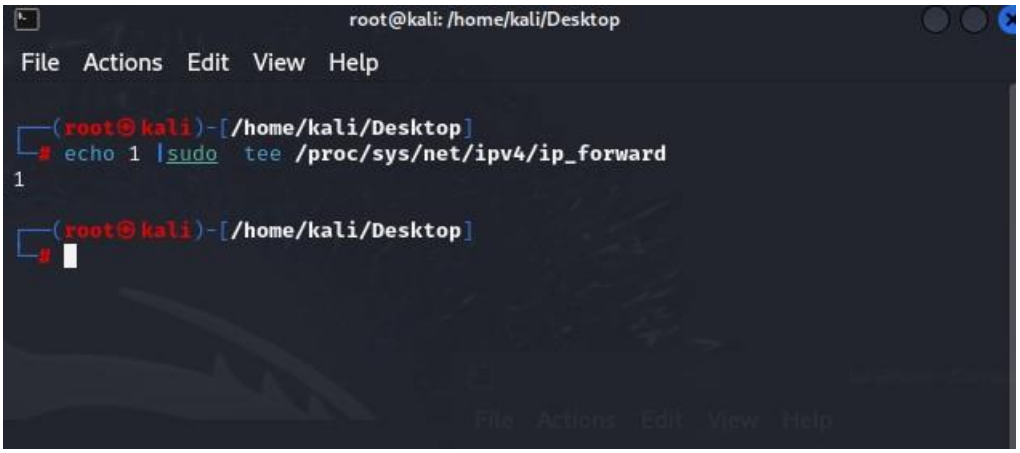

Figure 1: System Commands

1.3 **Network setup**

Start Bettercap with the following command ***"sudo bettercap"***. Make sure all the devices are connected to the personal hotspot. Once that is done, follow the commands from figure 2 to setup the network. Once all the IP addresses are scanned, choose the device to be targeted in my case, I chose my phone whose IP address is **10.197.201.196** (fig 2).

Type in the following commands to enable ARP Spoofing –

- ***set arp.spoof.targets 10.197.201.196***
- ***arp.spoof on***

Now open up Wireshark and start analysis. Filter the protocol by typing in DNS. Just to make sure make sure that the ARP spoof is working, search something from the target device. Monitor on wireshark if it is able to capture it. You can also do it by turning on sniff command in bettercap

```
File  Actions  Edit  View  Help
└─$ sudo bettercap
bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a li
st of commands]

10.197.201.0/24 > 10.197.201.74  » [21:50:21] [sys.log] [inf] gateway monitor
 started ...
10.197.201.0/24 > 10.197.201.74  » net.probe on
10.197.201.0/24 > 10.197.201.74  » [21:50:33] [sys.log] [inf] net.probe probi
ng 256 addresses on 10.197.201.0/24
10.197.201.0/24 > 10.197.201.74  » [21:50:33] [sys.log] [inf] net.probe start
ing net.recon as a requirement for net.probe
10.197.201.0/24 > 10.197.201.74  » [21:50:34] [endpoint.new] endpoint 10.197.
201.5 detected as 2c:3b:70:6f:1e:a1 (AzureWave Technology Inc.).
10.197.201.0/24 > 10.197.201.74  » net.show

┌──────────────┬──────────────────┬──────────┬──────────────────────────┐
│      IP ▲    │       MAC        │   Name   │          Vendor          │
Sent │ Recvd │   Seen   │
├──────────────┼──────────────────┼──────────┼──────────────────────────┤
│ 10.197.201.74  │ 00:0c:29:b5:1c:44 │ eth0     │ VMware, Inc.             │
0 B  │ 0 B   │ 21:50:21 │
│ 10.197.201.114 │ 8a:5f:22:e4:a3:fb │ gateway  │                          │
761 B │ 607 B │ 21:50:21 │
│              │                  │          │                          │
│ 10.197.201.5   │ 2c:3b:70:6f:1e:a1 │          │ AzureWave Technology Inc. │
0 B  │ 276 B │ 21:50:34 │
└──────────────┴──────────────────┴──────────┴──────────────────────────┘

↑ 37 kB / ↓ 89 kB / 1948 pkts

10.197.201.0/24 > 10.197.201.74  » [21:50:59] [endpoint.new] endpoint 10.197.
201.196 detected as f6:dd:28:46:5b:28.
10.197.201.0/24 > 10.197.201.74  » set arp.spoof.targets 10.197.201.196
10.197.201.0/24 > 10.197.201.74  » arp.spoof on
10.197.201.0/24 > 10.197.201.74  » [21:53:21] [sys.log] [inf] arp.spoof arp s
poofer started, probing 1 targets.
```

Figure 2: Network Setup

I searched tiktok.com and monitored on wireshark (figure 3).

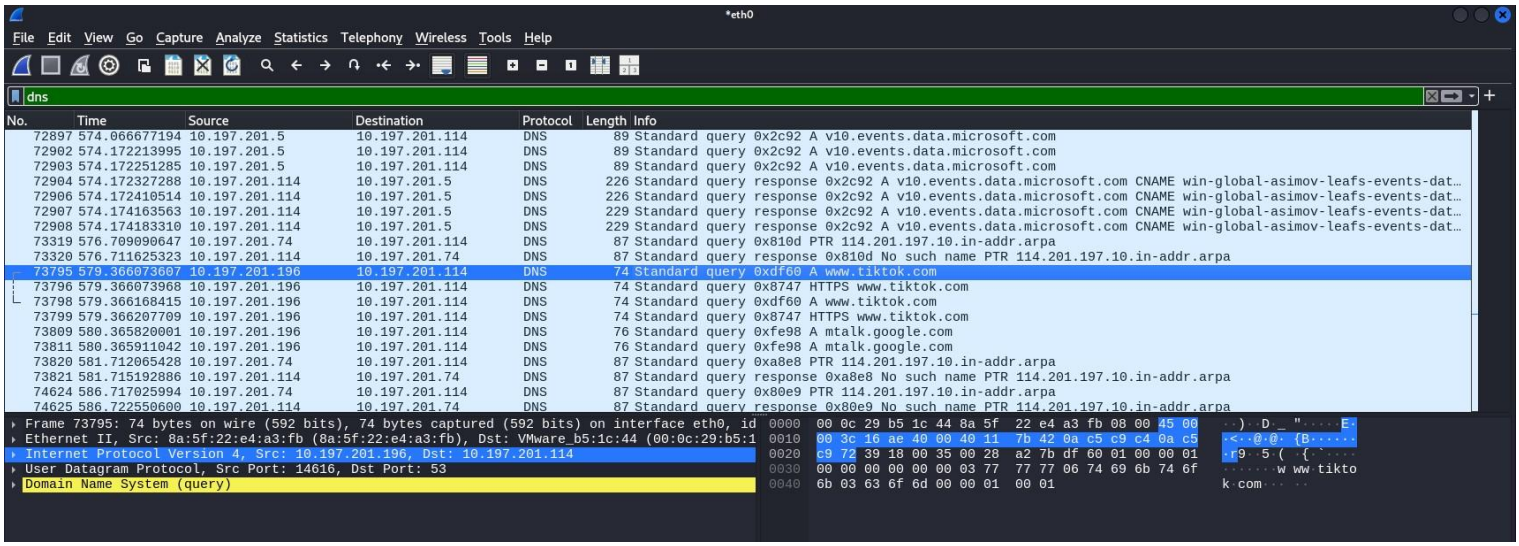

Figure 3: ARP Spoof Check on Wireshark

1.4 **Enabling DNS Spoofing**

Open a new terminal and type the following command

- ***sudo nano /usr/share/bettercap/caplets/dns.spoof***

This will open the DNS spoofing caplet for editing. Add the domain and IP address you want to spoof in the text file (figure 4).

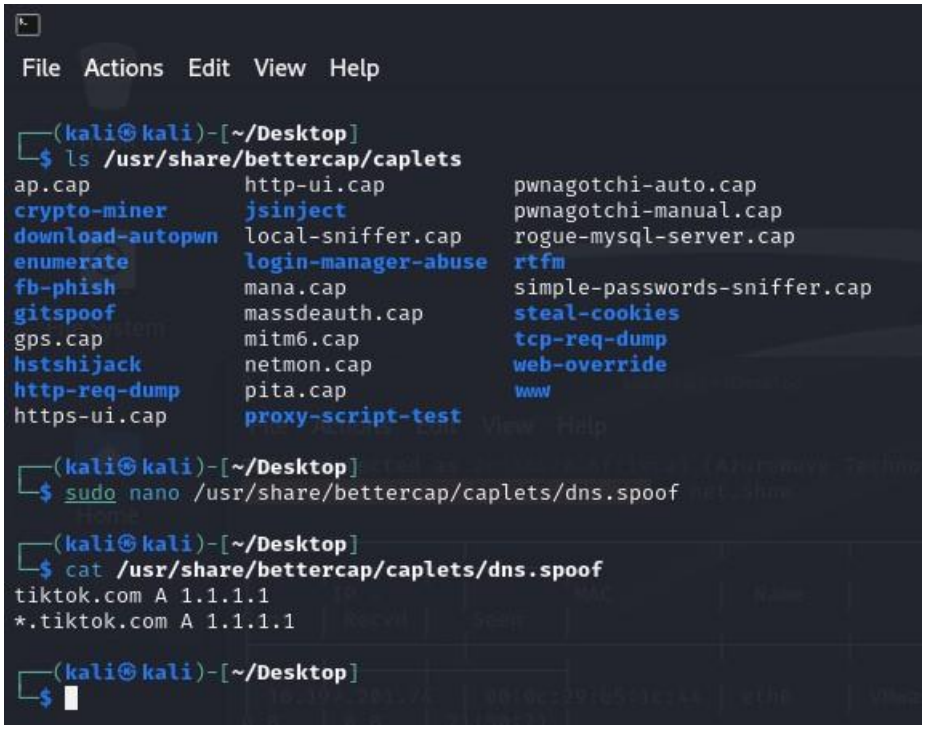

Figure 4: DNS Spoof Commands

In my case, I wanted to spoof tiktok.com to 1.1.1.1. Save and exit the file (Ctrl + O, Enter, Ctrl + X). Enable the DNS spoofing module in Bettercap (figure 5).

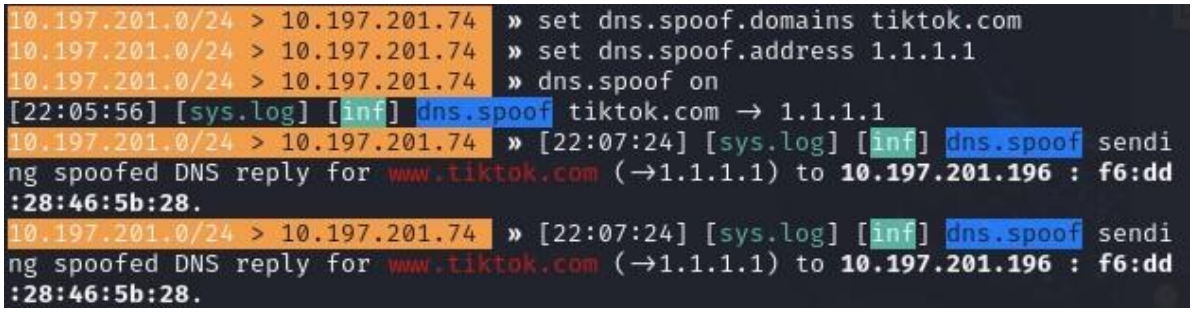

Figure 5: DNS Spoofing in Bettercap

1.5 **Monitoring the attack**

- On the target device, access the spoofed domain (e.g., tiktok.com) via a browser.
- Check Wireshark for DNS query responses showing the spoofed IP.
- If DNS spoofing works, the target device will either fail to load the site or be redirected to the spoofed IP.

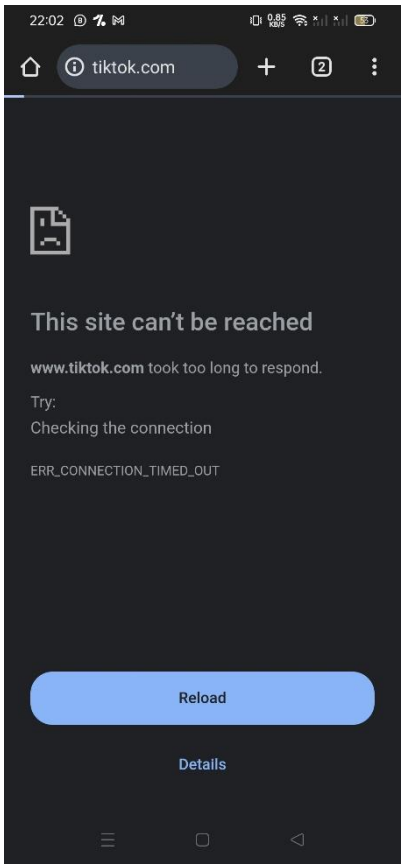

Figure 6: Search result from the target device

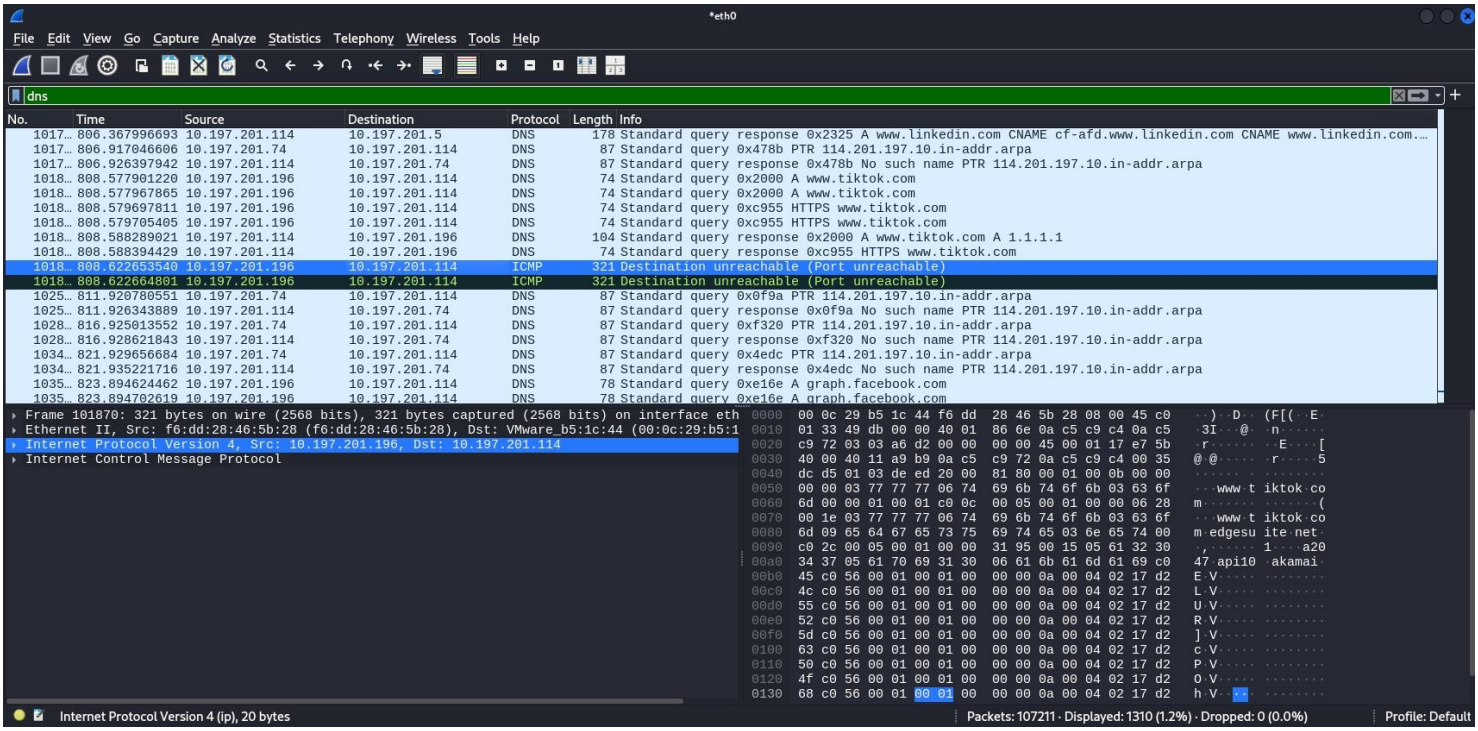

Figure 7: DNS Spoof Monitoring on Wireshark

Lessons Learned:

This project provided hands-on experience with:

- The mechanics of ARP and DNS spoofing.
- Using tools like **Bettercap** and **Wireshark** to simulate and monitor attacks.
- Understanding the importance of securing DNS infrastructure and mitigating MITM attacks through:
    a. Using encrypted DNS (e.g., DNS over HTTPS).
    b. Employing static ARP entries or network segmentation.

Conclusion:

This simulation highlights how attackers can exploit insecure network configurations to intercept or redirect traffic. By understanding these methods, we can better protect systems against MITM attacks.

This project is an educational exercise conducted in a controlled environment. Always obtain permission before testing on real networks.

References

Get Kali. (n.d.). Kali Linux. Retrieved 8 January 2025, from https://www.kali.org/get-kali/

Masas, R., Sharadin, G., Omri, Daniel, Muly Levy, Yerushalmi, S., & Threat, I. (n.d.). *ARP spoofing*. Learning Center; Imperva Inc. Retrieved 16 January 2025, from https://www.imperva.com/learn/application-security/arp-spoofing/

Tunggal, A. T. (2025). *What is DNS spoofing?* Upguard.com. https://www.upguard.com/blog/dns-spoofing